

Lemma F for the octahedron: opposite petals never overlap

working notes

September 2026 — draft

Setting and conventions

We use the conventions of `HANDOFF.md`. P is a convex polytope of octahedron type with antipodal (non-adjacent) vertices v, w and equator $u_0, \dots, u_3$ (the neighbours of w , indices mod 4, in counter-clockwise order around w seen from outside). The faces are $W_i = wu_iu_{i+1}$ and $V_i = vu_iu_{i+1}$; κ_x is the curvature at x ; ω_i, ν_i are the angles of W_i at w and of V_i at v . The net Z_k cuts $\text{star}(v) \cup \{wu_k\}$: the fan $W_k, W_{k+1}, W_{k+2}, W'_{k-1}$ around w (angular gap κ_w at the slit) with the petal V_i glued to W_i along u_iu_{i+1} . Touching counts as non-overlapping.

Only nine face pairs of Z_k share no net vertex: the three local pairs at the slit, the two opposite-petal pairs $(V_k, V_{k+2}), (V_{k+1}, V'_{k-1})$, and four petal-vs-far-fan pairs. Lemma F is the statement that under the hypothesis

$$\textbf{(H)} \quad \kappa_v \geq \kappa_x \quad \text{for every vertex } x \text{ of } P \quad (1)$$

none of the six far pairs overlaps, in every Z_k . (The handoff's hypothesis $\kappa_v \geq \max_i \kappa_{u_i}$ is weaker; the numerics below show that the weaker hypothesis is *not* enough for the argument of §3, although no counterexample to the weaker statement itself is known.)

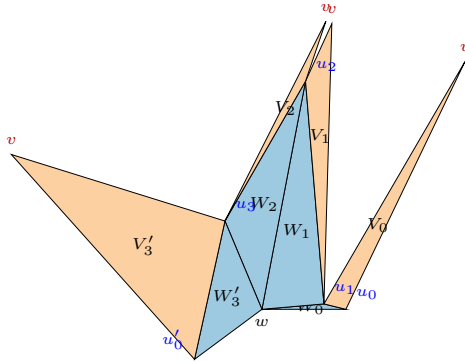


Figure 1: A typical net Z_k (fan faces blue, petals orange), v the sharpest vertex, slit at the sharpest neighbour u_0 of w .

Elementary consequences of (H). Since $\sum_x \kappa_x = 4\pi$ over six vertices, (H) gives $\kappa_v \geq 2\pi/3$, hence $\sum_i \nu_i = 2\pi - \kappa_v \leq 4\pi/3$. At any convex vertex each face angle is at most the sum of the other face angles (the spherical polygon inequality for the tangent cone), so $\nu_j \leq \frac{1}{2} \sum_i \nu_i$, i.e.

$$\nu_j \leq \pi - \frac{1}{2} \kappa_v \leq \frac{2\pi}{3} \quad \text{for every } j. \quad (2)$$

1 Reduction to three consecutive petals

Lemma 1 (petals do not cross interior hinges). *In Z_k no petal meets the relative interior of a hinge segment wu_j , $j \neq k$.*

Proof. The two fan faces adjacent to wu_j are W_{j-1} and W_j ; a petal crossing the segment would enter both. Every petal shares a net vertex with W_{j-1} or W_j (V_{j-1}, V_j share u_j ; V_{j+1} shares u_{j+1} with W_j ; V_{j-2} shares u_{j-1} with W_{j-1}), and faces sharing a net vertex lie in disjoint wedges there. $\square$

Lemma 2 (petal versus far fan face). *If a petal V_i overlaps a fan face W_j with $j \notin \{i-1, i, i+1\}$ then V_i overlaps V_j .*

Proof. V_i contains no vertex of W_j (w and the equator vertices of W_j lie in fan faces or petals that share a vertex with V_i , except possibly the local face W'_{k-1} , whose treatment is part of Lemma L). By Lemma 1 the boundary of V_i cannot enter W_j through its hinges, so it enters through the outer edge $u_j u_{j+1}$, which is the common edge of W_j and V_j ; near that crossing V_i meets V_j . $\square$

Hence Lemma F follows from the following statement about three consecutive petals, applied to the triples (V_k, V_{k+1}, V_{k+2}) and $(V_{k+1}, V_{k+2}, V'_{k-1})$ of Z_k .

Conjecture 3 (Triple Lemma). *Let W_i, W_j, W_{j+1} ($j = i+1$) be three consecutive fan faces of a net and V_i, V_j, V_{j+1} their petals. Under (H), V_i and V_{j+1} do not overlap.*

Numerical evidence 4. No overlap of opposite petals under (H) in 72,000 random nets, nor under the weaker hypothesis $\kappa_v \geq \max \kappa_u$; adversarial hill-climbing (edges ≥ 0.05 diam, curvatures ≥ 0.05) brings the two petals to within 0.4% of the diameter but never into overlap. All observed overlaps (44 in 72,000 nets, without hypothesis) have two equator vertices sharper than v .

2 Where a meeting can be

Fix a triple as in Conjecture 3 and write $u = u_j$, $u' = u_{j+1}$ for the base of V_j , ϕ, ψ for the base angles of V_j at u, u' , $\kappa = \kappa_u + \kappa_{u'}$. In the net, V_i and V_j meet at u only, separated there by the angular gap κ_u between the two copies uv_j and uv_i of the cut edge uv ; likewise at u' . Let L be the line through uv_i and L' the line through $u'v_{j+1}$.

Lemma 5 (double-outward wedge). *V_i lies in the closed half-plane of L not containing V_j , and V_{j+1} in the closed half-plane of L' not containing V_j . Hence every common point of V_i and V_{j+1} lies in $\mathcal{D} :=$ the intersection of these two half-planes, which is the wedge at the point $p^* = L \cap L'$ opposite to the base uu' . If $\kappa < \nu_j$ then $\mathcal{D}$ lies on v_j 's side of the base line (above), and if $\kappa > \nu_j$ it lies on w 's side (below); $\mathcal{D} = \emptyset$ if $\kappa = \nu_j$.*

Proof. The rays uv_i and $u'v_{j+1}$ make angles $\phi + \kappa_u$ and $\psi + \kappa_{u'}$ with the base, so they converge above the base iff $\phi + \psi + \kappa < \pi$, i.e. $\kappa < \nu_j$; the rest is the description of the four wedges at p^* . $\square$

Lemma 6 (the two cut edges never cross). *The segments uv_i and $u'v_{j+1}$ are disjoint.*

Proof. If they crossed at p then p would be above the base with base angles larger than ϕ, ψ , so v_j lies inside the triangle $uu'p$ and $|up| + |u'p| > |uv_j| + |u'v_j| = |uv_i| + |u'v_{j+1}|$, contradicting $|up| \leq |uv_i|$, $|u'p| \leq |u'v_{j+1}|$. $\square$

Claim 7 (observed overlaps). *In the 44 observed opposite-petal overlaps (no hypothesis), the intersection region lies in $\mathcal{D}$; 26 are above the base and 18 below, exactly as predicted by the sign of $\nu_j - \kappa$. In every below-base case the outer edges uu_i and $u'u_{j+2}$ of W_i, W_{j+1} converge on w 's side (equivalently $\angle_u W_i + \angle_u W_j + \angle_{u'} W_j + \angle_{u'} W_{j+1} < \pi$) and exactly one of the two fan faces extends past the crossing point; both cannot, since fan faces are disjoint.*

3 The rotation picture (above-base case)

Glue V_i, V_j, V_{j+1} flat around v with no gaps (the “ v -fan”); this is possible because $\nu_i + \nu_j + \nu_{j+1} < 2\pi$, and in it the three petals occupy disjoint wedges at v . The net position of V_i is obtained from its v -fan position by the rotation ρ_u about u by the gap angle κ_u (away from V_j), and that of V_{j+1} by the rotation $\rho_{u'}$ about u' by $\kappa_{u'}$. Both rotations have the same sense. Therefore

$$V_i \cap V_{j+1} \neq \emptyset \text{ in the net} \iff \mathcal{R}(V_i^{\text{fan}}) \cap V_{j+1}^{\text{fan}} \neq \emptyset, \quad \mathcal{R} := \rho_{u'}^{-1} \circ \rho_u,$$

and $\mathcal{R}$ is the rotation by $\kappa = \kappa_u + \kappa_{u'}$ about the point c^* on w 's side of the base with $\angle c^*uu' = \kappa_u/2$, $\angle uu'c^* = \kappa_{u'}/2$ (composition of two rotations). Let M be the line c^*v ; $\mathcal{R}$ moves points from u' 's side of M towards u 's side. Measure angles around c^* from M , positively towards u 's side.

Lemma 8 (angular criterion). *Let $\varepsilon_i \geq 0$ be the largest angle (seen from c^*) by which $V_i^{\text{fan}} \cap \{\text{above the base}\}$ crosses M towards u' 's side, and ε_{j+1} the corresponding excursion of $V_{j+1}^{\text{fan}} \cap \{\text{above}\}$ towards u 's side. If $\kappa < \nu_j$ and*

$$\varepsilon_i + \varepsilon_{j+1} < \kappa, \tag{3}$$

then V_i and V_{j+1} do not overlap in the net.

Proof. By Lemma 5 a common point p is above the base. The rotation ρ_u^{-1} maps the wedge of V_i at u in the net, $[\phi + \kappa_u, 2\pi - \angle_u W_j - \angle_u W_i]$, onto the v -fan wedge, so $x = \rho_u^{-1}(p)$ is below the base only if p is; hence $x \in V_i^{\text{fan}}$ and $y = \rho_{u'}(p) \in V_{j+1}^{\text{fan}}$ are above the base and $y = \mathcal{R}(x)$. Their angles from M satisfy $\theta_y = \theta_x + \kappa$ with $\theta_x \geq -\varepsilon_i$ and $\theta_y \leq \varepsilon_{j+1}$, so $\kappa \leq \varepsilon_i + \varepsilon_{j+1}$ (no wrap-around since $\kappa < \nu_j < \pi$). $\square$

In particular, if neither petal crosses M above the base ($\varepsilon_i = \varepsilon_{j+1} = 0$), there is no overlap. When does a petal cross M ? Write $\mu_u = \angle uvc^*$, $\mu_{u'} = \angle c^*vu'$, so $\mu_u + \mu_{u'} = \nu_j$ (c^* lies in the cone of V_j at v , being on w 's side of the base between the lines vu, vu'). The wedge of V_i at v starts at vu and turns away from V_j by ν_i ; it reaches the ray of M beyond v iff $\nu_i > \pi - \mu_u$. The other way for $V_i^{\text{fan}} \cap \{\text{above}\}$ to cross M is through the far vertex u_i or through the point where the edge vu_i or uu_i crosses the base line.

Lemma 9. *Under (H) the two petals cannot both cross M through their wedges at v .*

Proof. That would need $\nu_i + \mu_u > \pi$ and $\nu_{j+1} + \mu_{u'} > \pi$, hence $\nu_i + \nu_{j+1} + \nu_j > 2\pi$, whereas $\nu_i + \nu_j + \nu_{j+1} \leq \sum \nu \leq 4\pi/3$. $\square$

Lemma 10 (the above-base case needs a very sharp v). *Under (H), $\kappa < \nu_j$ forces $\kappa_v > 6\pi/7$, hence $\sum_i \nu_i < 8\pi/7$ and each $\nu_i < 4\pi/7$.*

Proof. $4\pi = \sum_x \kappa_x \leq 4\kappa_v + \kappa$ and $\kappa < \nu_j \leq \pi - \kappa_v/2$ by (2). $\square$

Numerical evidence 11. In 3000 random octahedra (all apexes, all triples with $\kappa < \nu_j$): the composition $\mathcal{R}$ has the stated centre and sense, and the criterion of Lemma 8 reproduces the net overlap test exactly. Under (H) neither petal ever crosses M above the base ($\varepsilon_i = \varepsilon_{j+1} = 0$ in all 224 triples, degenerate ones included), and adversarial search under (H) cannot make either excursion positive. Under the weaker hypothesis $\kappa_v \geq \max \kappa_u$ a petal can cross M and (3) can fail (excursion 1.96 with $\kappa = 0.05$), so (H) is genuinely used. Every one of the 26 observed above-base overlaps violates $\varepsilon_i = \varepsilon_{j+1} = 0$ (20 through V_i , 6 through V_{j+1}).

What remains for the above-base case. Prove that under (H) and $\kappa < \nu_j$ neither $V_i^{\text{fan}} \cap \{\text{above}\}$ nor $V_{j+1}^{\text{fan}} \cap \{\text{above}\}$ crosses M (or, more weakly, that (3) holds). The three crossing mechanisms are: (a) the wedge at v : $\nu_i > \pi - \mu_u$; (b) the far vertex u_i above the base on u 's side of M ; (c) an edge of V_i crossing the base line outside the segment uu' on u 's side of M . Mechanism (c) through vu_i requires $\nu_i + \nu_j > \pi + \psi$; (a) requires $\nu_i + \nu_j > \pi$; by Lemma 10 these need $\kappa_v < \pi$ and two of the four angles at v close to $4\pi/7$. The numerics say the fan faces then obstruct this, but the argument is not yet written.

4 The below-base case

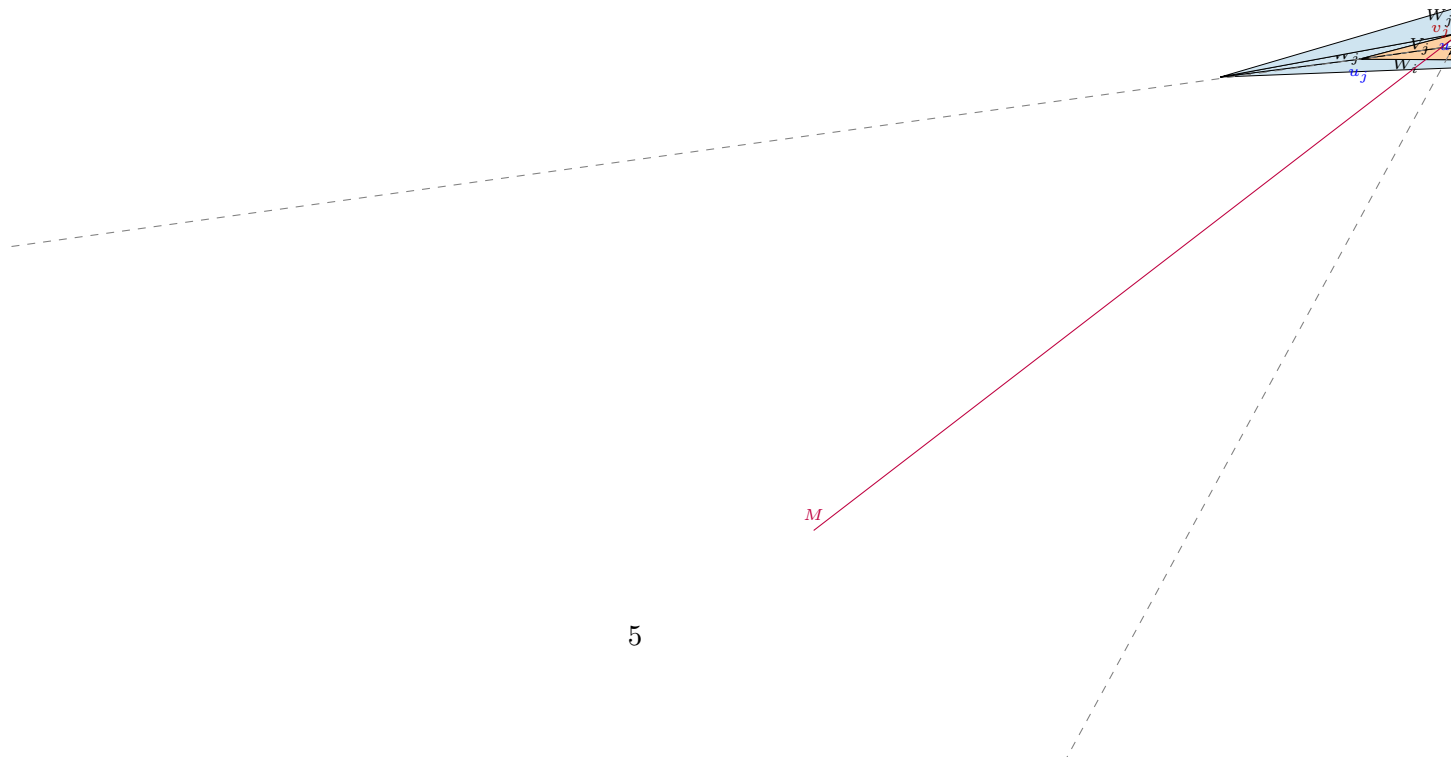
When $\kappa > \nu_j$ the wedge $\mathcal{D}$ lies on w 's side of the base. Then (Claim above) a meeting requires the outer-edge lines of W_i and W_{j+1} to converge on w 's side, one of the two fan faces to reach past their crossing point X , and the opposite petal to pass over the far vertex of that fan face. In the observed below-base overlaps two equator vertices have curvature close to 2π and v is nearly flat. Under (H), $\kappa_u, \kappa_{u'} \leq \kappa_v$ and $\nu_j < \kappa \leq 2\kappa_v$; no proof yet.

5 Known false / do not retry

Simple separating lines for the two petals inside $\mathcal{D}$ (bisector of $\mathcal{D}$ at p^* , lines through v_j) fail in about a third of the non-degenerate cases even under (H): the separation is a rotation phenomenon, not a half-plane one. The six wedge-lean inequalities of the handoff fail in 0.5% of nets under (H) (needle-like w), so they cannot replace the argument above.

Code

`octa.py` (module: all quantities, nets Z_k , nine pairs, intersection polygons), `vwedge.py`, `oppcases.py` (collect/classify/draw overlaps), `adv_opp.py` (adversarial separation), `cstar.py` (checks of §3), `adv_eps.py`, `adv_mu.py`, `adv_cstar.py` (adversarial tests of (3) and of the crossing conditions), `tikz_net.py` (figures), `draw.py` (PNG).



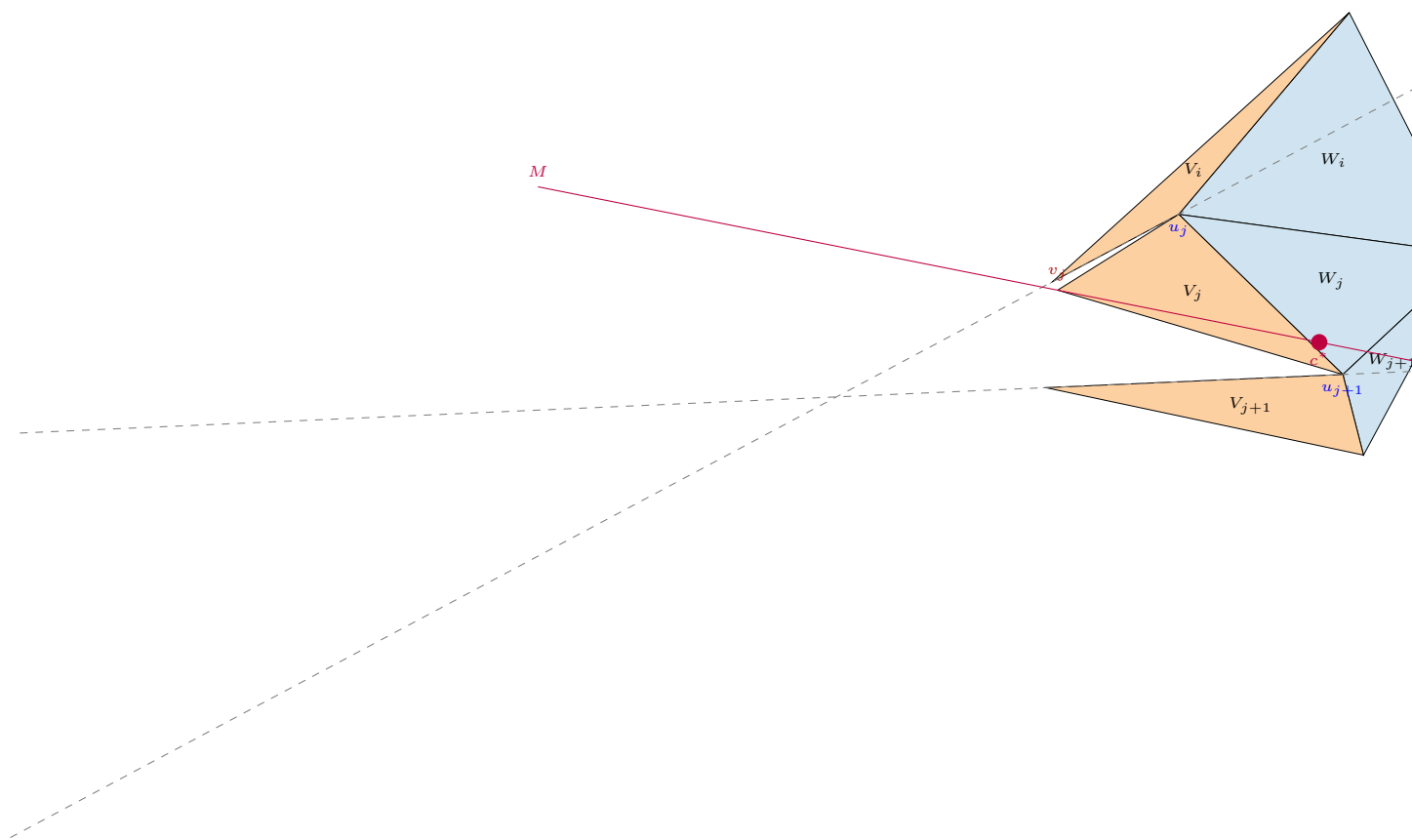


Figure 3: A triple under (H) with $\kappa < \nu_j$: c^* , the line M and the two dashed cut-edge lines; both petals stay on their own side of M .